Module – 4: Site plan, design, and Resource allocation

4.1 Understanding Customer Requirements

To understand customer's needs:

- Be clear with the methods that you will use to gather the required information.
- What specific questions you will answer with each of those methods?





Customer Requirements

1. How much power for loads ?
2. How much energy for backup ?
3. Where is the suitable space in the building for the components of a PV system ?

Methods for Gathering Customer Requirements

Assessment by Phone	• Customer's expectation from a solar system ? • Devices the customer uses? • How much power cuts?
Assessment by Walkthrough	• Devices and their ratings? • How much space is available for the system? • Risks and obstacles on site?
Reading Electricity Bills	• Units of electricity consumed? • How much is the user currently paying for electricity?
Predicting Shadows on the Space	• Will the available space be free from shadows throughout the year?

Assessment by Phone

Use the following sample questions while conducting an assessment from the phone:

1. What devices do they typically use?
2. Do they use any heavy load devices such as air-conditioners or pumps?
3. Do they need to power part or all of their loads?
4. What is their estimated load size?
5. How many hours of power cut do they have?
6. How many hours of backup do they need?
7. At what time of the day and months are power cuts more frequent?



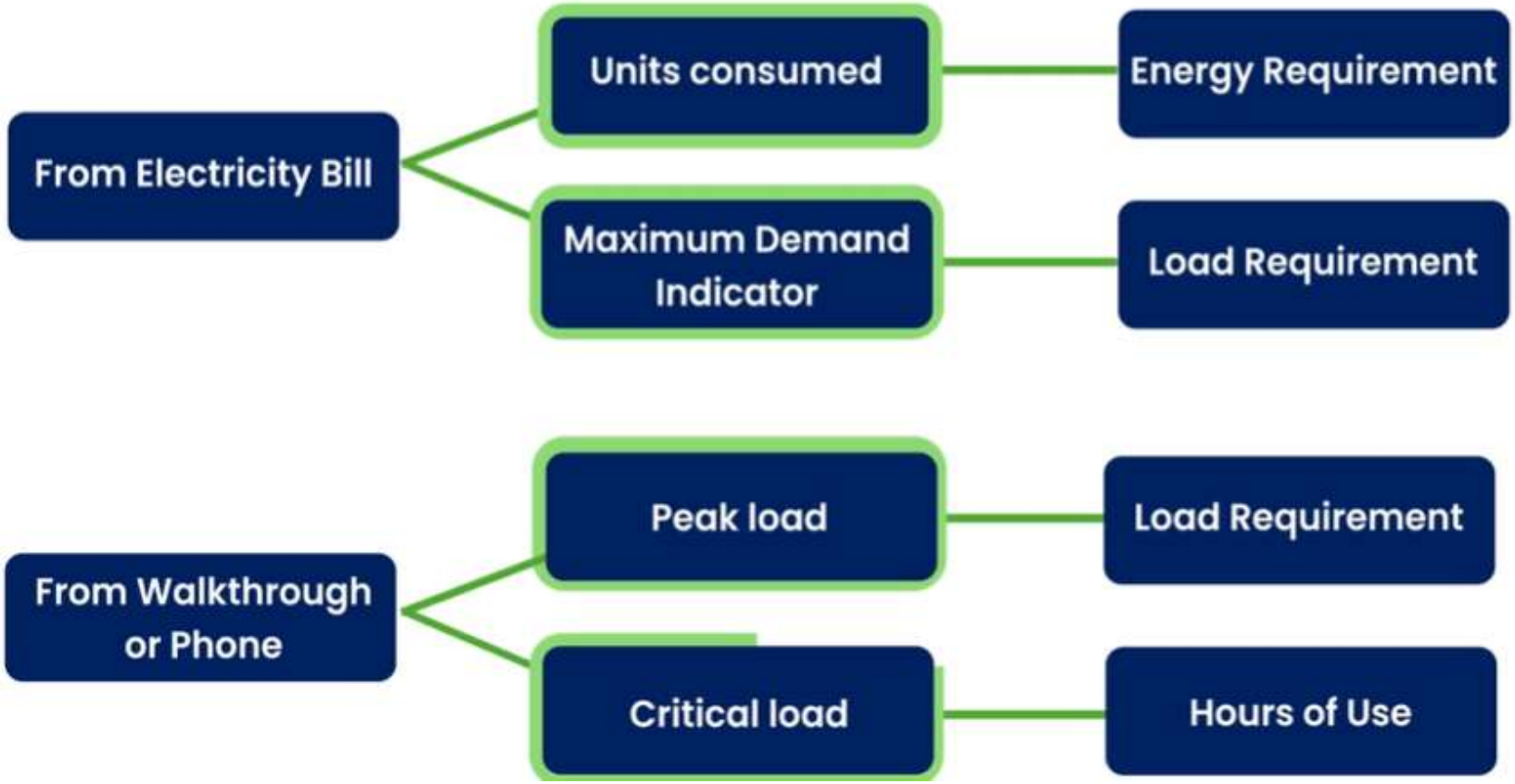
Assessment by Walkthrough





Visit site for a deeper and practical understanding

Summary of methods to do load and site assessment



Units consumed – kWh of energy consumed by devices

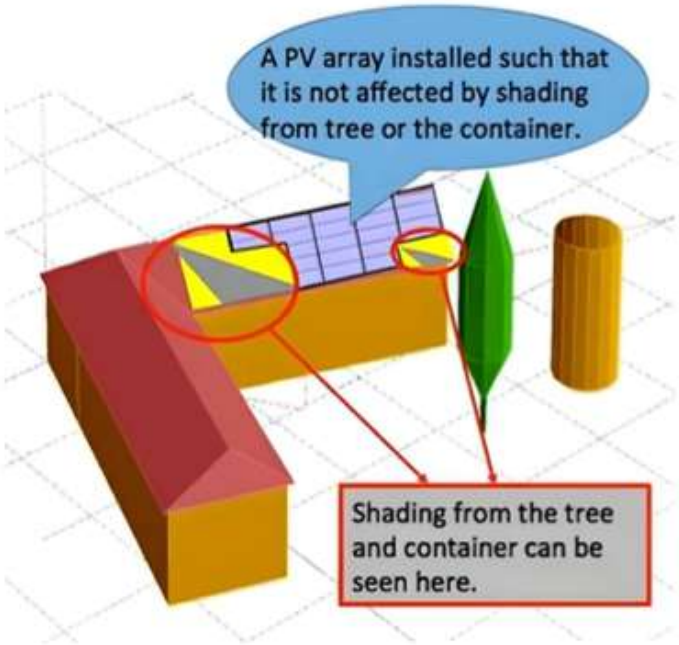
Maximum demand indicator - The maximum amount of electrical energy that is required by a specific household or a facility

Peak load - The maximum load that can occur when all devices are operated.

Critical load - Loads that are absolutely essential to work during an outage.



The angle of sunlight changes with time and seasons, impacting the energy received. The sun's position greatly affects solar energy systems.



Placement of PV array to avoid shading from surrounding objects

Remember, that the PV array should be installed in a such a manner that it is not affected by the shade of a tree or any other large object.



PV array installations carried out after thorough consideration of possible shading objects surrounding the location.

Accurate shading analysis allows for tailored system designs, ensuring cost-effectiveness for clients.



Site Survey Activities	Some benefits of doing an accurate assessment
Load Assessment	Solar array and batteries are sized according to need Critical loads are backed up The customer's get maximum value for money
Site Assessment	Components are placed in proper locations Shadows do not fall on the arrays maximising output The area remains safe for human and equipment

4.2: Steps for conducting load assessment

4.2.1 Gather Load of Information



Step-1: Gathering Details of the Customer







Question	Answer
1. Name and address of customer	
2. Type of customer	Resident/Commercial/Institutional
3. Type of Locality	City/Town/Village
4. Current sources of power used	Electricity Grid/Diesel Generator/ DC solar devices
5. Back - up • Current back-up sources used • Battery backup size , if any (in Ampere- hours)	 • Yes/No • Battery back up
6. If grid is available, hours of power cuts experienced	

Sample Form to Gather Details of Customer

Question	Answer
1. Name and address of customer	
2. Type of customer	Resident/Commercial/Institutional
3. Type of Locality	City/Town/Village
4. Current sources of power used	Electricity Grid/Diesel Generator/ DC solar devices
5. Back - up • Current back-up sources used • Battery backup size , if any (in Ampere- hours)	 • Yes/No • Battery back up
6. If grid is available, hours of power cuts experienced	



Keep in mind:

A residential customer is looking to install a system for his own household.

A Commercial customer includes shops, offices, and industries.

Examples of institutional customers will be schools, colleges, hospitals, and government buildings.

The form that seeks customer data will provide you an opportunity to know your customer better.

Step-2: Identify Devices and Their Ratings



Identify all the major electrical devices and record their ratings.



Peak load:maximum load powered by the solar system.



Sometimes, only certain critical loads are backed up using solar energy.

Type of Load	Power	Total Number	Critical loads	Hours of use
AC Loads				
TV	60 W	1	1	3
Fan	60 W	5	2	6
Lamp	60 W	3	0	6
CFL	20 W	12	6	6
Desktop Computer	270 W	2	1	4
Laptop	50 W	2	0	2
Laser Printer	375 W	1	0	
Refrigerator	380 W	1	0	12
DC Loads				
Phone Charger	4 W	1	1	5

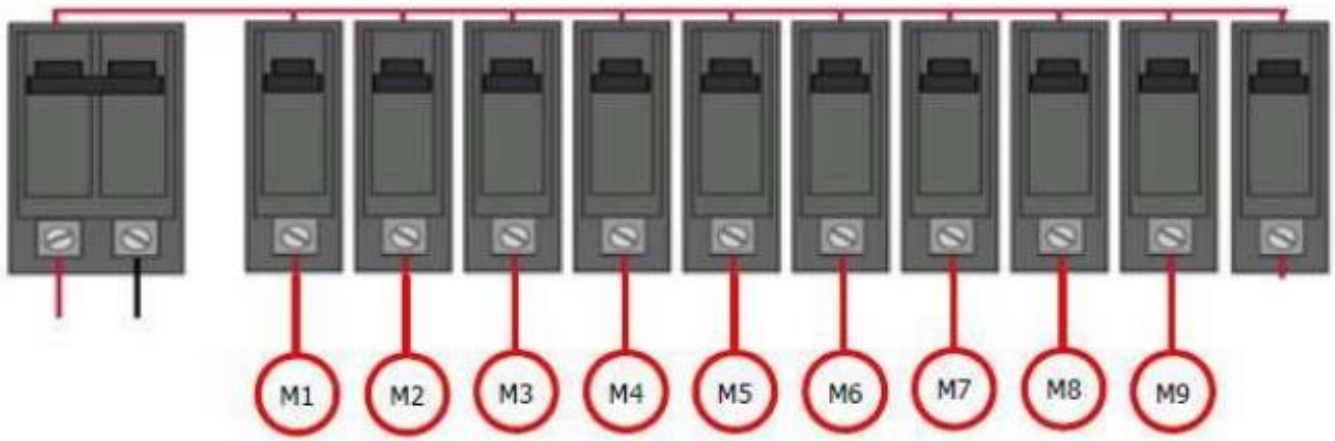
Identifying MCB Units

Hello Learners! Now we know how important MCB units are to us. Therefore, we should make it a point to identify and capture all normal and power loads.

In your home wiring layout, you can number all the MCBs installed at the site and then identify the loads controlled by them. This can be represented in a tabular form. This information is very useful while designing the PV system for the site.

Let's get started, identify all the MCBs and their respective loads and represent the data in a tabular form as shown below.

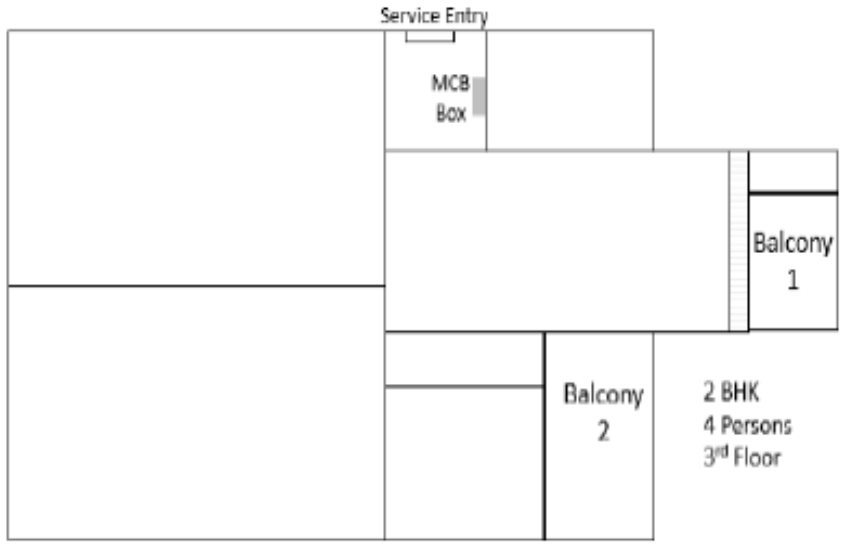




You can enter your observations like this:

MCB	Remarks	Electrical Load	Power Socket
M1	DR	TV, Incandescent Light Bulb	Yes (2)
M2	BR1		Yes (1)
M3	BR2	TV	Yes (1)
M4	Kitchen	Tube light, Fan, Laptops	Yes (3)
M5	Terrace	Light	Yes (1)

Now, Identify the service entry, and draw a basic layout of home wiring marking locations such as the Service entry panel.



Hope this was fun! Now you know everything about the steps to conduct a Load Assessment.



Keep in mind:

You can usually find the wattage of most appliances stamped on the bottom or back of the appliance, or in its nameplate.You can also find the detailed specification of equipment by downloading their datasheet or user manual.

Gather all information without making assumptions about the type of solution the customer will need.



Keep in mind:

Gather all information without making assumptions about the type of solution the customer will need.

If the wattage is not listed:

- Estimate it by finding the current draw (in amperes) and multiplying that by the voltage used by the appliance.
- Most appliances in India use 240 volts.
- The amperes might be stamped on the unit in place of wattage. If not, find a clamp-on ammeter –Take a reading while the device is running; this is the actual amount of current being used at that instant.
- The initial current drawn by motor will be high.

Step-3: Read the Electricity Bill

1. Take note of the bills of last 12 months.
2. Record the units consumed each month.
3. Record the unit electricity charges mentioned in the bill.
4. Note the additional surcharges applied by the distribution company.



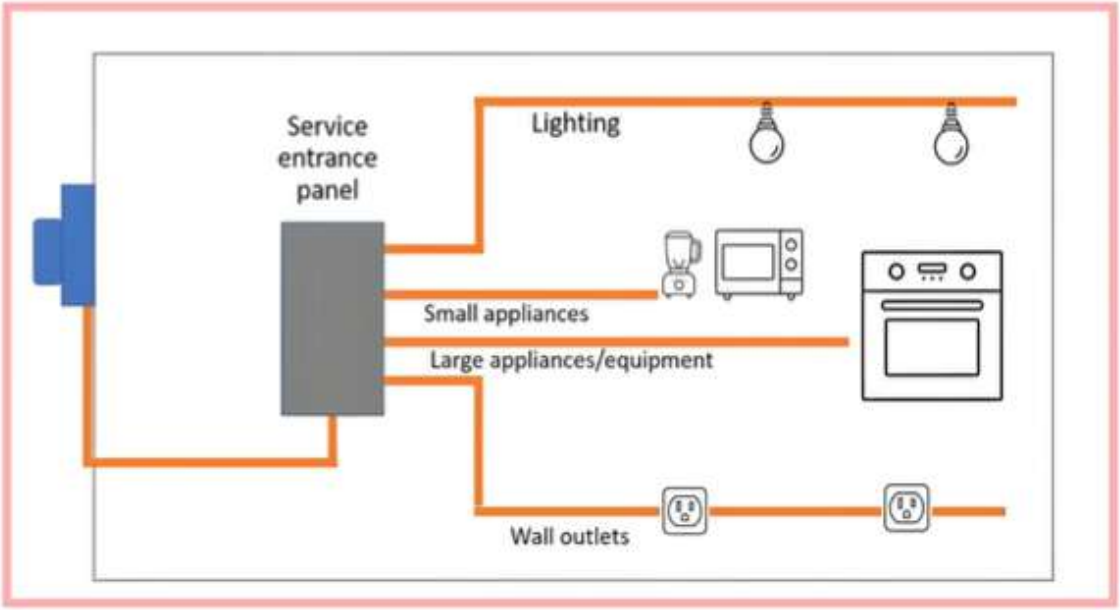
Month	Units Consumed	Unit Rate Applied	Surcharges	Month	Units Consumed	Unit Rate Applied	Surcharges
Jan				July			
Feb				Aug			
Mar				Sep			
April				Oct			
May				Nov			
June				Dec			



Information recorded from Electricity Bill:

Branch circuit

Refers to the isolated areas of the home where the panel board directs electrical current



Important to understand how each branch circuit operates.



To give details regarding the wiring of the house/building.

Miniature Circuit Breaker (MCB)

It is a specific type of overcurrent protection device used in a branch circuit.

It's the reason one can cut power to one room while leaving the power of the rest of the house operational.



Working of MCB Unit:

MCB has reusable automatic switch that interrupts the flow of current when it exceeds a predetermined safe limit.

When MCB detects current overload, it automatically trips and protects the wires and connected devices.



4.3: Steps for Conducting Site assessment

Step-1: Prepare Layout of the Building

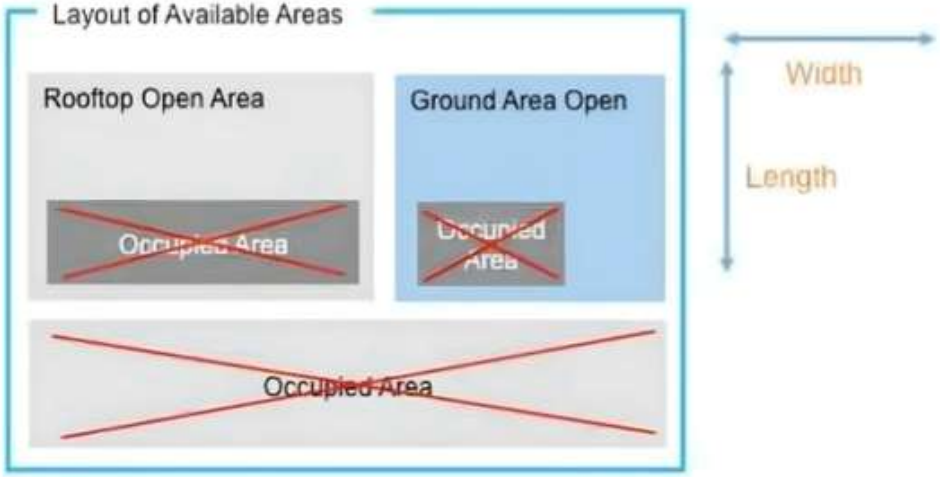


- Layout preparation.
- Mark places available with shading for batteries and inverters
- Mark nearest points for MCB/ Electricity meters.



Step-2: Prepare Layout of Roof; Take Pictures; Mark Dimensions

- Walk through the terrace and draw the layout.
- Mark the dimensions and directions.

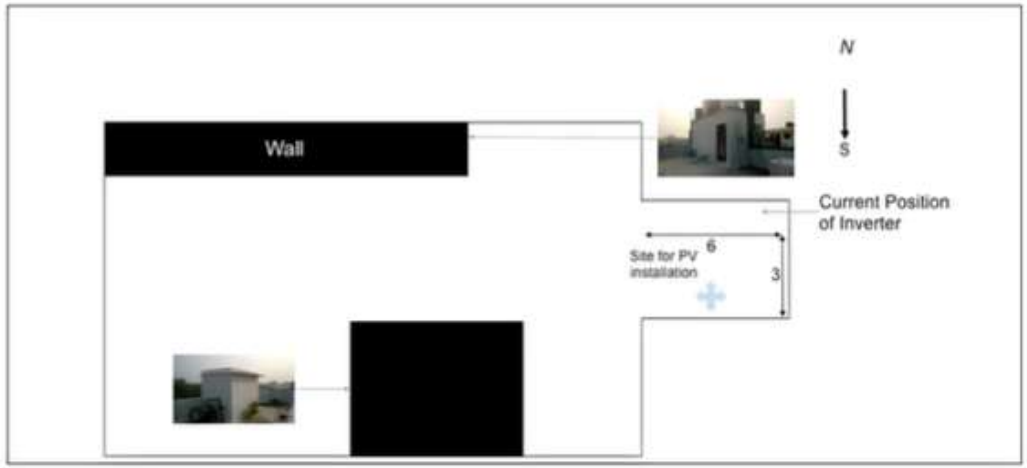




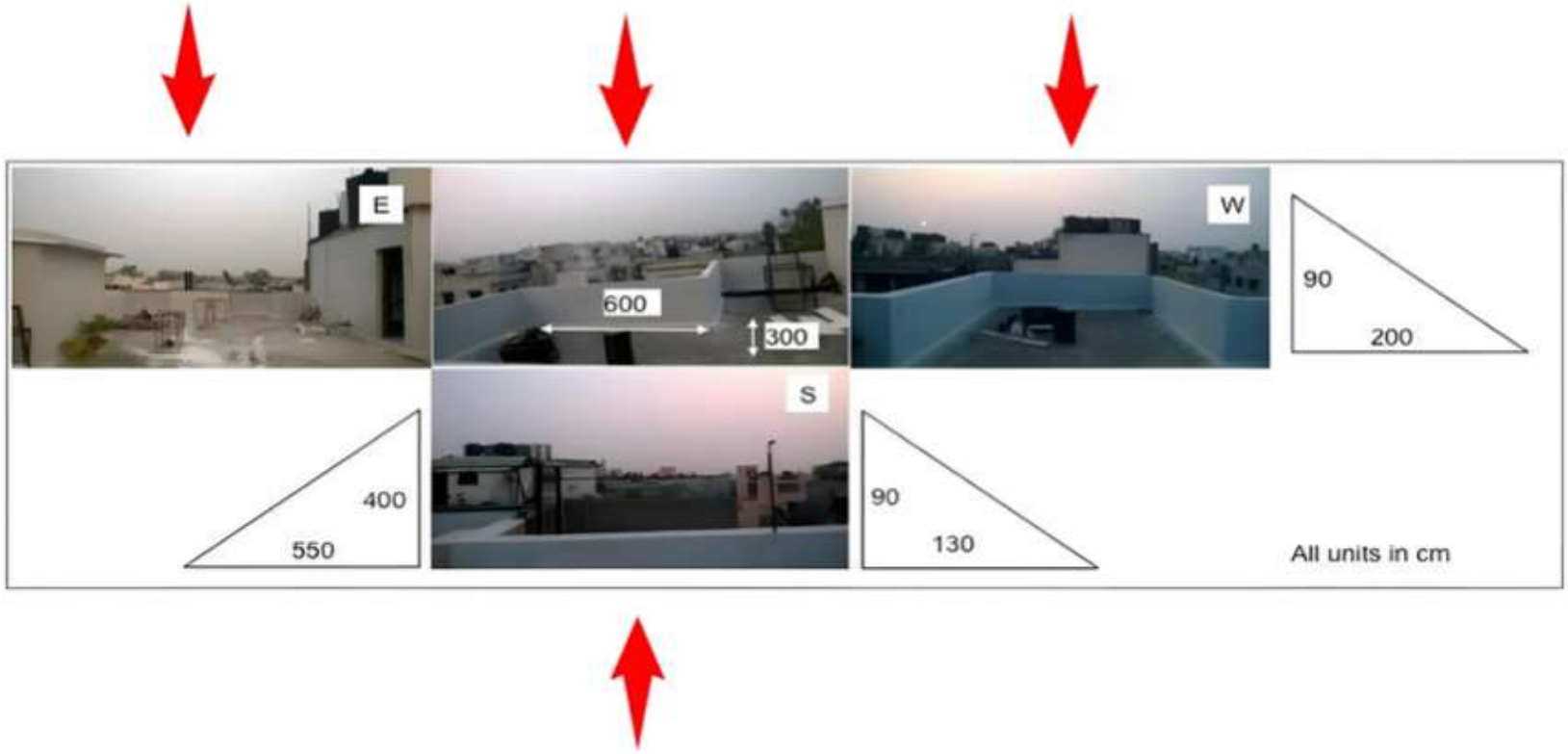
- Mark the visible obstructions.
- Enquire whether any of these obstructions can be moved or adjusted to optimize the installation area.



- Multiple photographs are taken from all angles to document the roof's current state.
- Identify and mark the most suitable spots on the roof for installing the PV array, ensuring these areas are optimal for maximum sunlight exposure.






E
600
300
W
90
200
S
400
550
90
130
All units in cm
N
S

Use of Google Earth Pro Software









Google earth pro imagecfor the location having coordinates 28°22'8.17"N 76°55'52.32"E

Use of Google Earth Pro Software

Utilised in Solar Power Site Surveys	Advanced Imaging and Resolution- Provides high-resolution satellite imagery, which is crucial for detailed analysis and visualisation of potential solar power plant locations.
Detailed Terrain and 3-D Visualisation	Allows users to assess the terrain with greater accuracy. The 3D visualisation capabilities are essential for understanding how specific land features might affect the placement and efficiency of solar panels

Enhanced Measurement Tools	Offers more sophisticated tools for measuring area, distance, and radius, which are essential for layout planning and infrastructure development within a solar power plant.
Sunlight and Shadow Simulation	Includes tools to calculate the sun's trajectory at different times of the year, which helps in estimating potential solar power generation and analysing shading effects from surrounding objects.

Layering and Data Importing	Users can import GIS data, including land use, vegetation cover, and topographic maps, which can be overlaid on the satellite images to provide a multi-dimensional view of the site.
Historical Imagery	Provides access to historical satellite images, allowing users to observe changes over time in the chosen site area, which can be critical for long-term planning and compliance with environmental regulations.

Advanced Printing and Export Options

Enables high-quality printing and the ability to export images and data in various formats, facilitating the sharing of findings and proposals with stakeholders.



Limitations of Google Earth Pro Software:

Accuracy concerns: Despite its enhanced capabilities, the accuracy of software's imagery and data might still require validation through physical site surveys.

Data Currency: The satellite imagery in Google Earth Pro might not reflect very recent developments or changes in the landscape.



Keep in mind:

Inverters should be in a room or shelter protected from sun and rain.

Batteries should be in a ventilated room or shelter on a rack.

If no such space exists, ask the customer about constructing shelters for the equipment.

Inverters should be as close as possible to the DC combiner box to minimise DC side wire loss.

Draw a rough sketch of the cable routing and estimate the wiring length needed up to the switchboard.

Step-3:Choose Spots Suitable for Shading Analysis

- Check if it's possible to easily carry the panels to the area.
- Ensure the spot is as close to the DC combiner box and inverter. This proximity helps minimize power losses in the DC wiring.

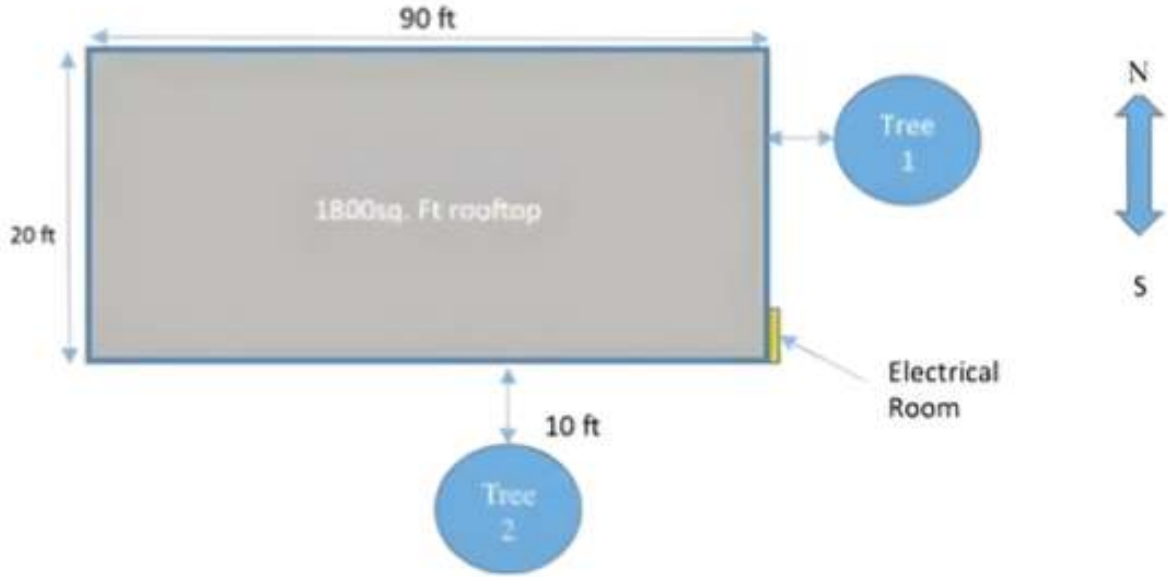


Step-3:Choose Spots Suitable for Shading Analysis



- Assess the area required for the solar array, looking for shadow-free locations suitable for installation and nearest to the inverter wiring.
- Roughly 100 square feet or 10 square meters of area is required per kilowatt.
- Nowadays 80 sq feet or 8 sq meters of are is required for a 1 kW solar plant installation. Solar panel comes with a higher rating of 500-550 Wp.





Sample layout diagram of a rooftop

- Shading situations present a significant challenge in preparing PV system implementation plans, as shading can greatly affect system yield.

- Shading analysis tells us how many hours of sunlight a selected location will receive in a given month of the year.



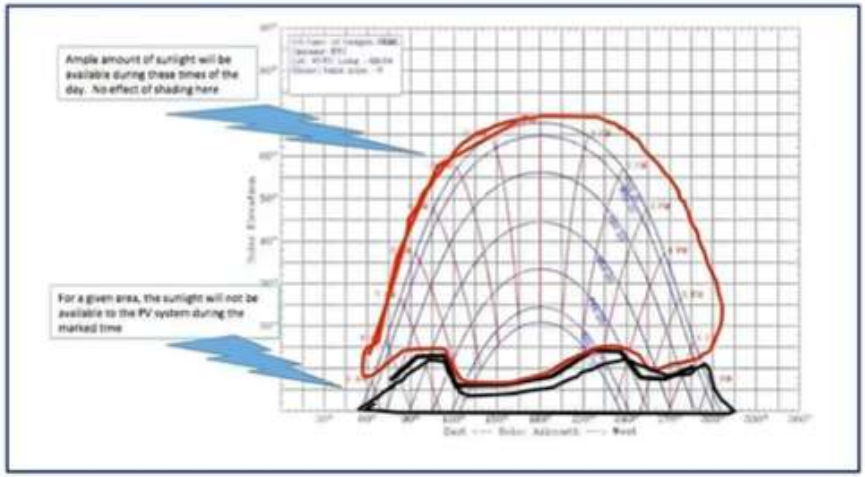
- Simulation programs, such as **Sketchup**, **PVSOL** and **PVsyst,** use 3D horizon photographs to model shading effects and project annual energy generation.

- Tools like Solar Pathfinder, Sun Eye, and paper tools can also perform shading analysis.

- Data from these tools can be input into simulation software for accurate energy generation projections.



Step-4: Performing Shading Analysis



Sunlight availability mapped on a sun path diagram during the entire day

Website: https://solardata.uoregon.edu/SunPathChart.html



3D modelling of rooftop solar PV system



Detailed shading analysis- Effect of shading on individual solar module.

4.4: Delivering PV solutions from customer requirements

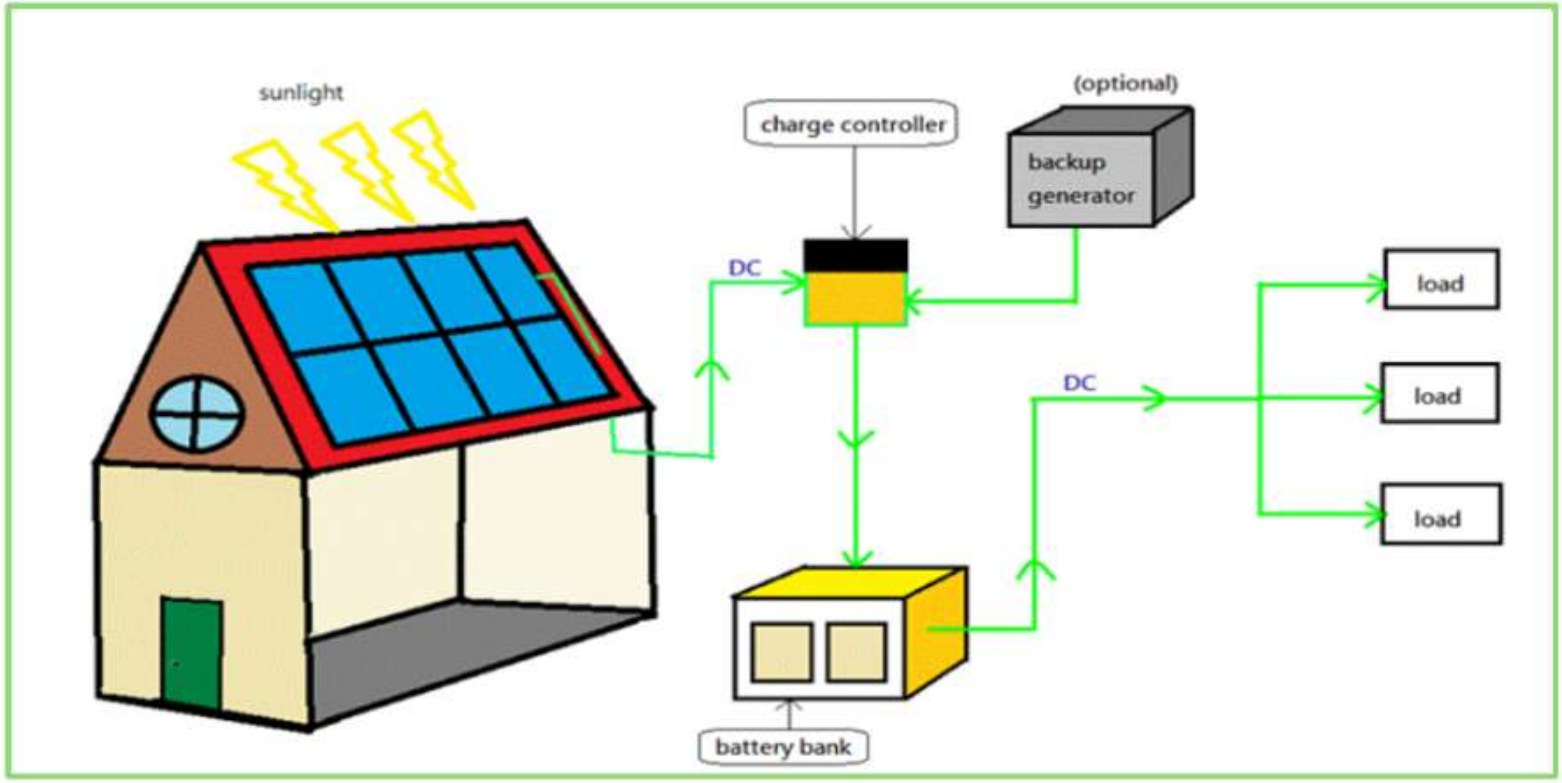
4.4.1 Factors Deciding the Type of Solar PV Systems





DC Off-Grid PV System

- Operates independently of the electric grid.
- Utilises solar panels to convert sunlight.
- Relies on batteries to store excess solar energy.

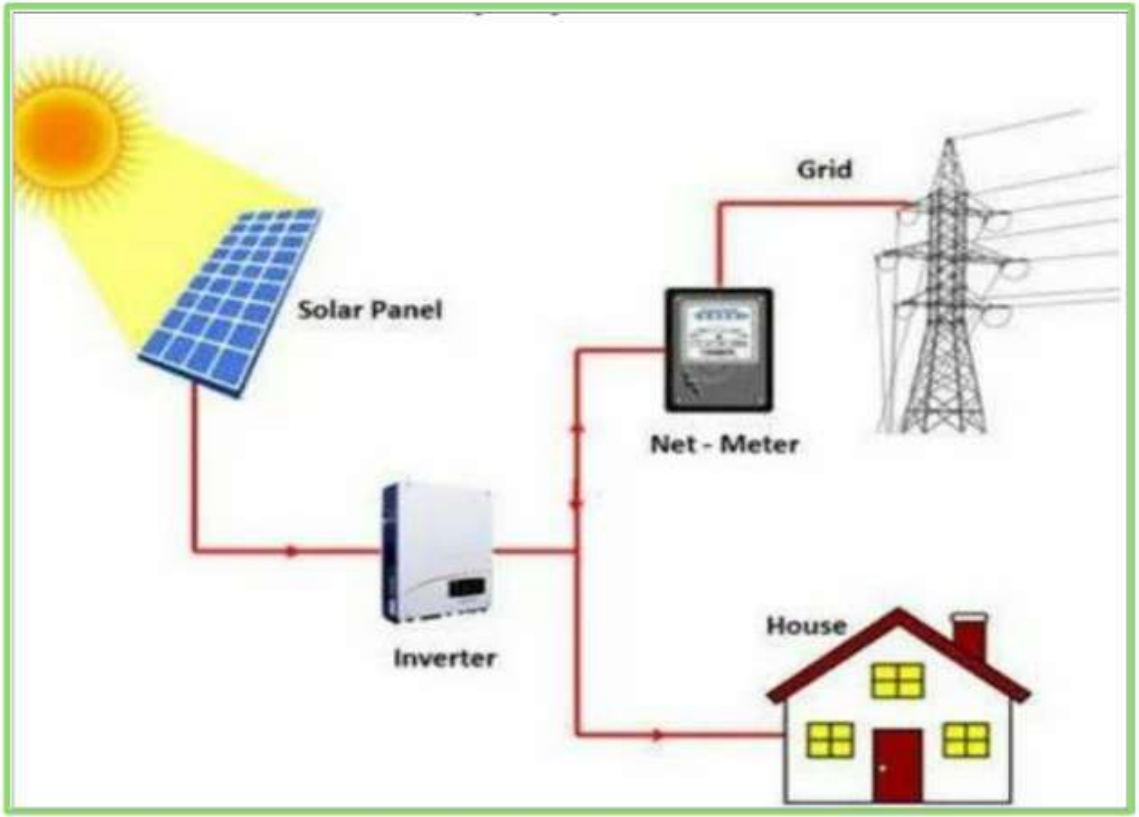




- Utilises solar panels to convert sunlight.
- No connection to electrical grid.
- Relies on batteries to store excess solar energy.

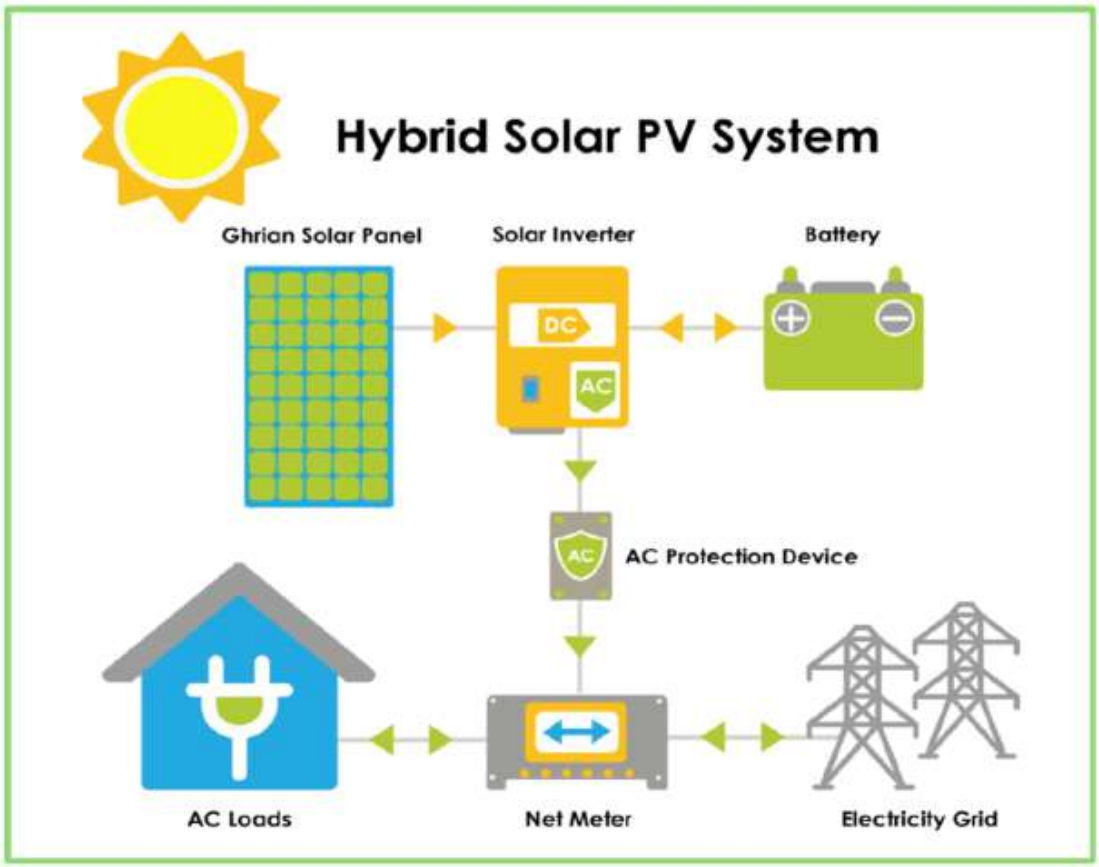
Grid- tied Solar PV System

- Most commonly used.
- Converts sunlight to DC Electricity.
- Connected to a utility grid.
- During sunshine, used for self-consumption.
- Excess power generated is fed back into the utility grid.



Hybrid Solar PV System

- Has solar power generation, battery storage, and grid connection.
- Converts sunlight to DC Electricity.
- Battery bank stores excess solar energy generated which can be used in the absence of sunlight.
- Connected to the utility grid for electricity backup.



Differences between DC off-grid and AC Off Grid Solar PV System is:

DC Off Grid

DC off-grid systems use DC appliances directly.

AC Off Grid

AC off-grid systems primarily focus on powering AC appliances.

Factors Deciding the Type of Solar PV System



Customer Requirement	PV System Type
• Lightening, mobile charging, small devices such as DC fans	DC off-grid system
• AC loads such as fans, CFLs, TV and Refrigerators • Very erratic or no grid supply. So the backup requirement is high.	AC off-grid system (stand-alone system)
• Grid is available with very little or no power cuts	Grid-tied system
• Grid is available and need 24-hour power	Hybrid Solar System

4.5: Prepare Bill of Materials (BOM)

Composition of Solar PV Systems



Solar Charge Controller

- Regulates the Voltage and Current from the PV Panels.
- Prevents battery overcharging and deep discharging.



PV Module

A photovoltaic (PV) module, also commonly known as a solar panel, is a flat-plate device that converts sunlight into DC electricity using the photoelectric effect.



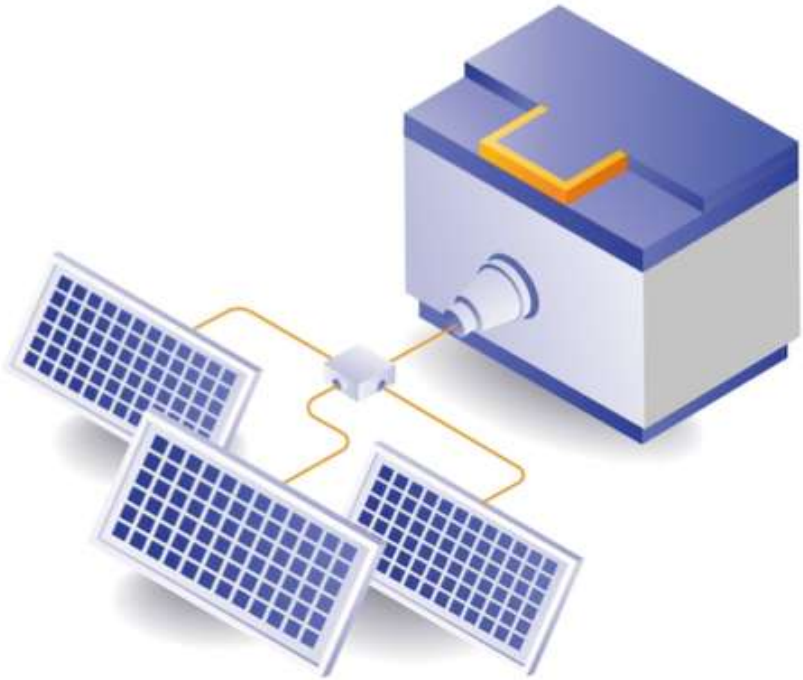
Inverter

Power electrical device that converts DC electricity into an AC electricity.



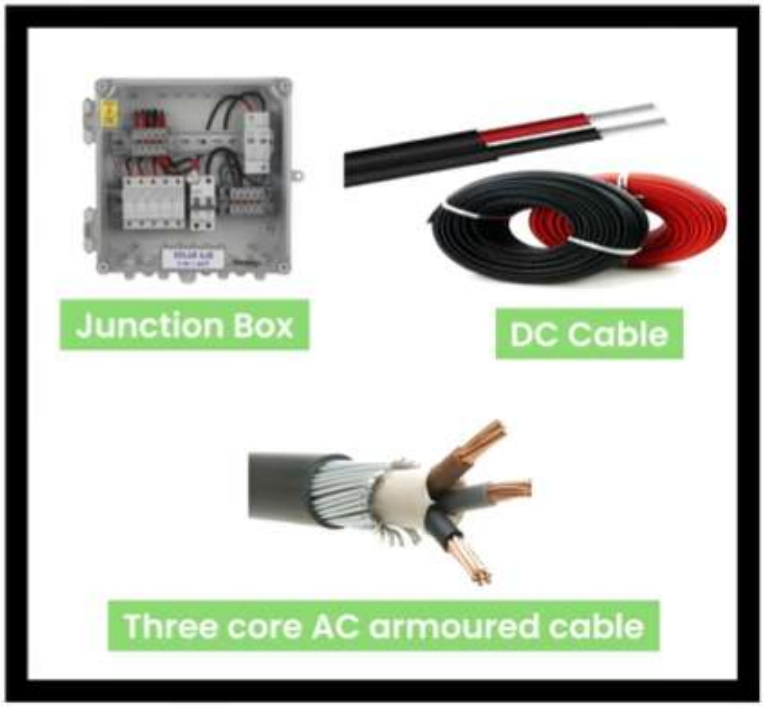
Cables (AC and DC)

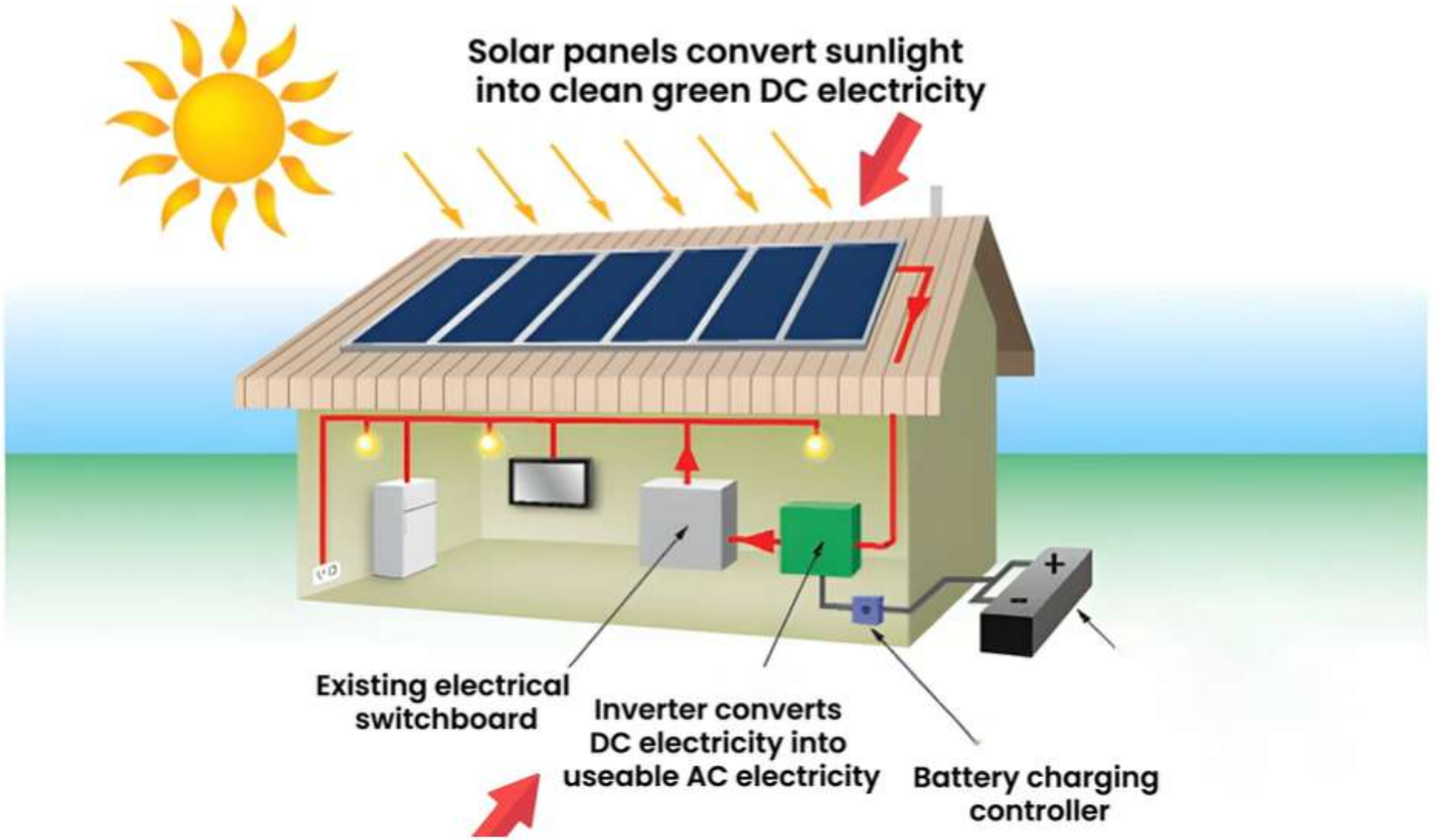
Stores energy for electrical appliances when there is demand.

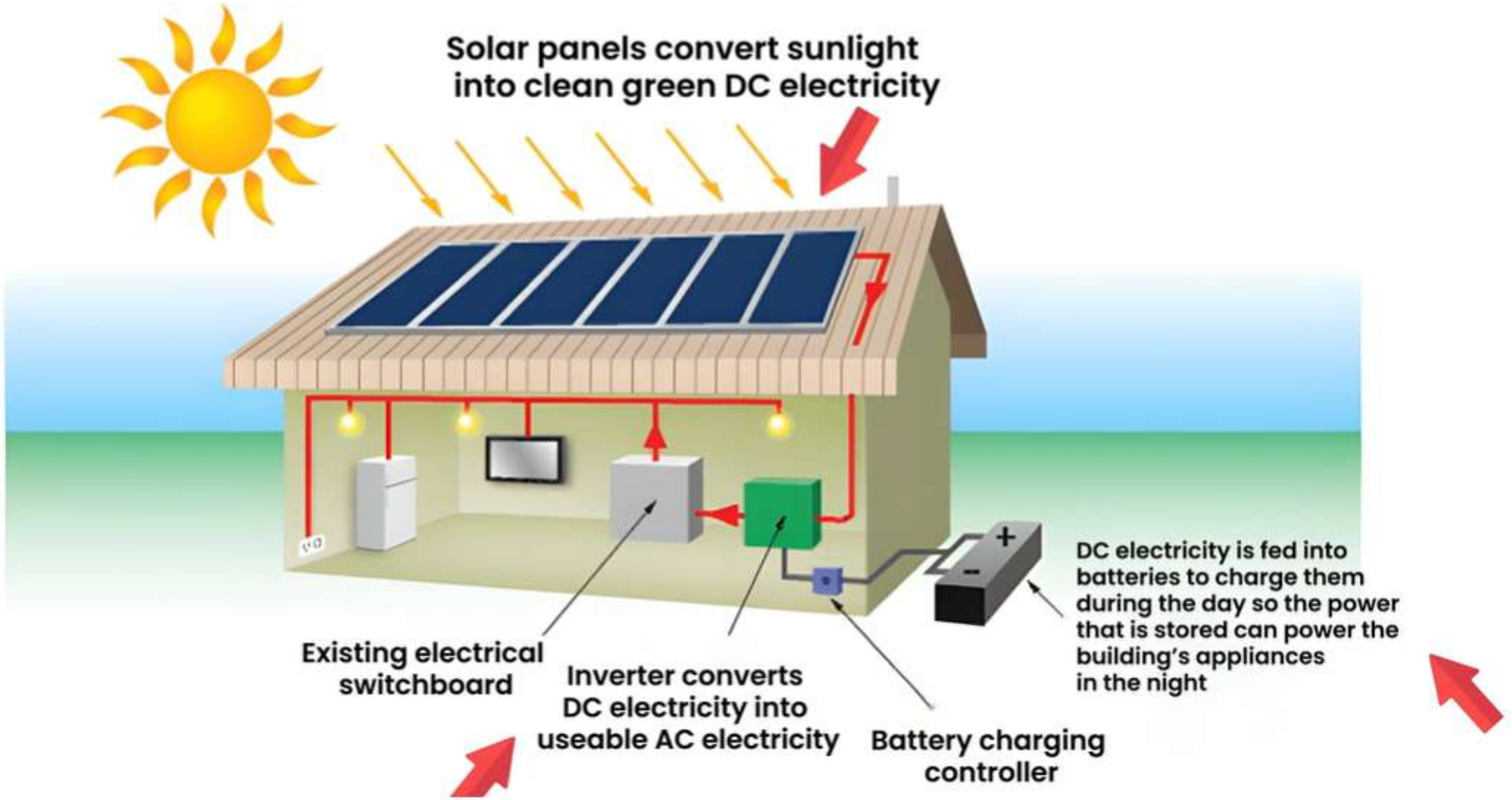


Other Accesories

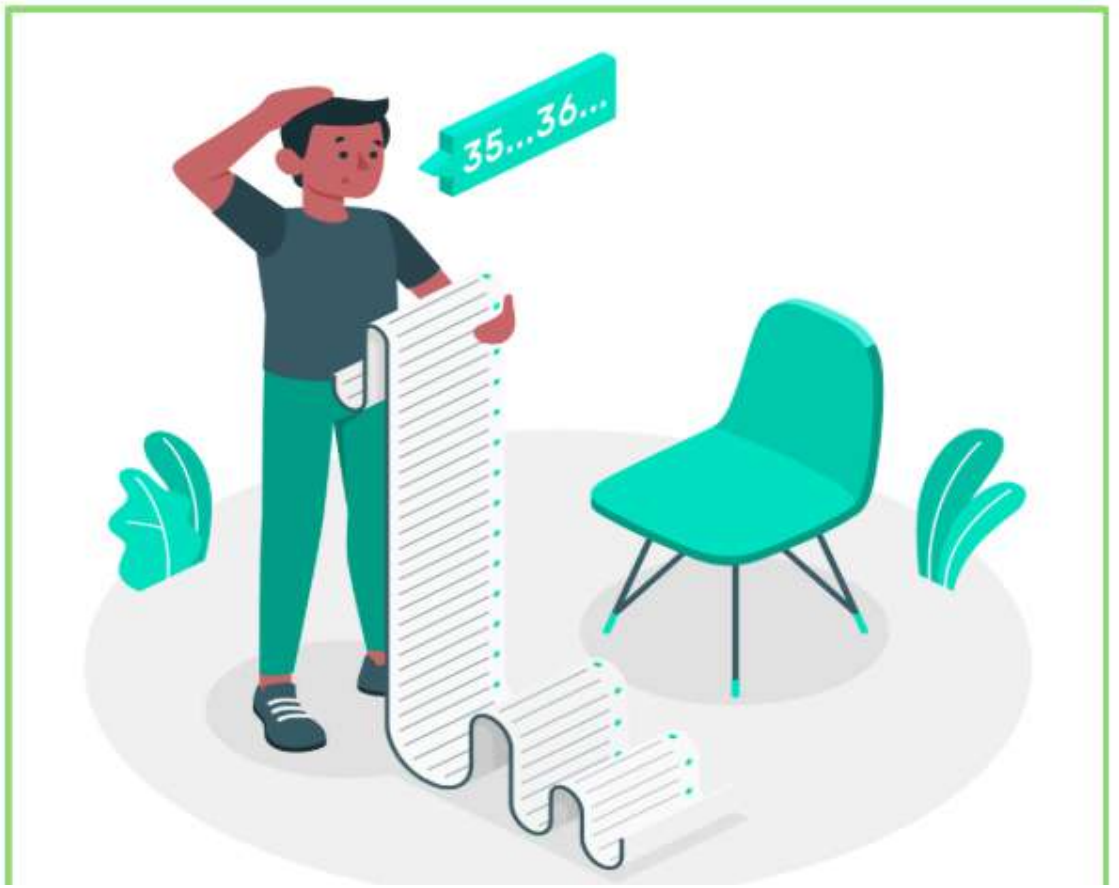
Some of the other yet important accessories include cable wires and junction box.




Solar panels convert sunlight into clean green DC electricity
Existing electrical switchboard
Inverter converts DC electricity into useable AC electricity
Battery charging controller
+
-


Solar panels convert sunlight into clean green DC electricity
+
-
DC electricity is fed into batteries to charge them during the day so the power that is stored can power the building's appliances in the night
Existing electrical switchboard
Inverter converts DC electricity into useable AC electricity
Battery charging controller

4.5.2 Bill of Material



Bill of Material (BOM)

A formally structured list of objects.It lists all the components of the object.



Components of a Bill of Material



Unit Measure of Each Component

Quantity

Reference Number

Object With Name

Types of Bill of Materials



Engineering Bills of Material (EBOM)



Manufacturing Bills of Material (MBOM)

Difference between EBOM and MBOM



Thus, engineering and manufacturing usually have different valid views of the same product.

Format of Bill of Material

Bill Of Materials					
Sl.No.	**Item**	**Description**	**Quantity**	**Cost**	**Remarks**

Prepare BOM from Single Line diagram

Single Line Diagram (SLD):

- A Single Line diagram is also called a One-line diagram.
- It is a simplified notation for representing an electrical system.

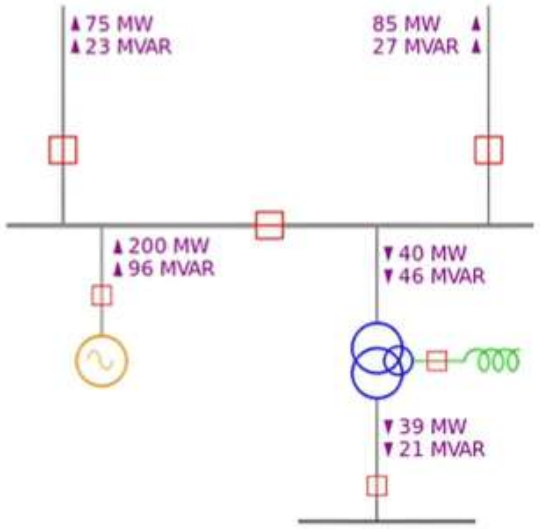
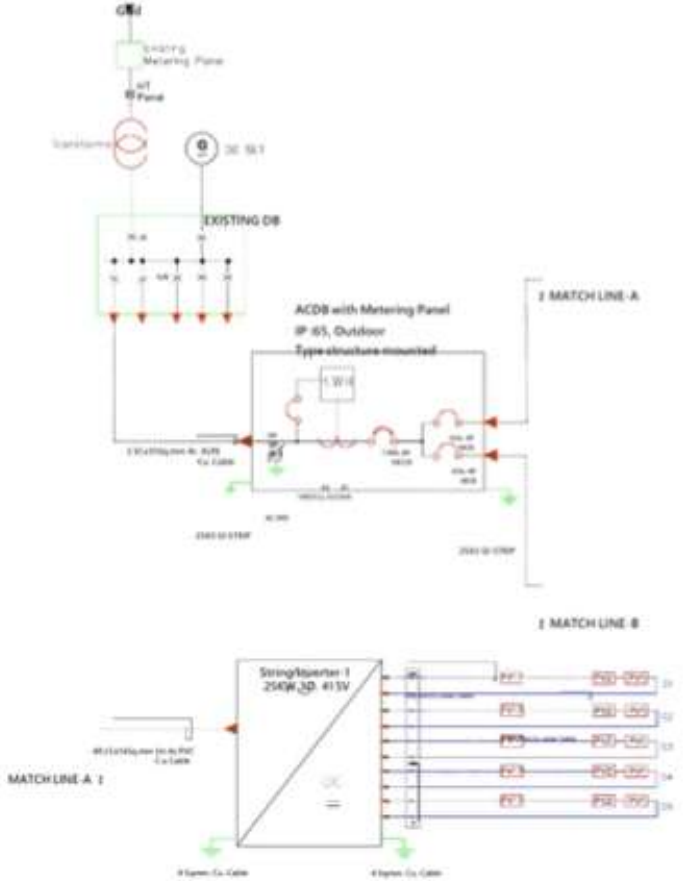
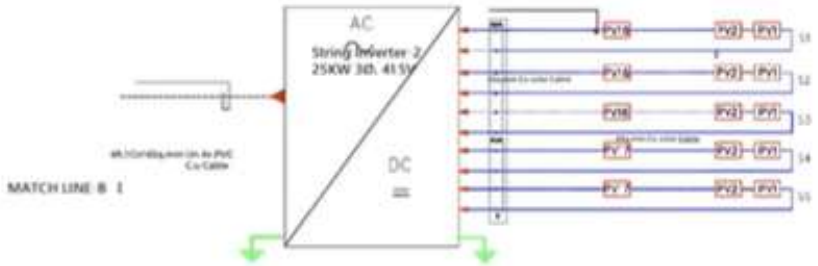


Image source: https://en.wikipedia.org/wiki/Single-line_diagram

SINGLE-LINE DIAGRAM





Legends

S.No.	Symbol	Descriptor
1.		Modules
2.		String Inverter
3.		MCB
4.		MCCB
5.		Circuit Breaker
6.		CT
7.		AC SPD



Let's Quantify,

SAMPLE BILL OF MATERIAL				
ELECTRICAL COMPONENTS				
SL.	ITEM	DESCRIPTION	QTY	
1	Modules	300 wp, As per Specification	167	
2	Inverter	25 KW Capacity, as per specification	2	
3	ACDB Metering Panel	IP65, outdoor type structure mounted	1	
4	Cable	1C X 16 SQ. MM.Cu, AR, PVC	AS PER REQUIRED	
5	Grounding Wire	4SQ.MM, Cu	AS PER REQUIRED	
6	Tools and Tackles	AS PER VENDOR RECOMMENDATION		
7	Consumables		LOT	

Prepare BOM from Civil/ Structure Drawings

Civil/ Structure Drawings

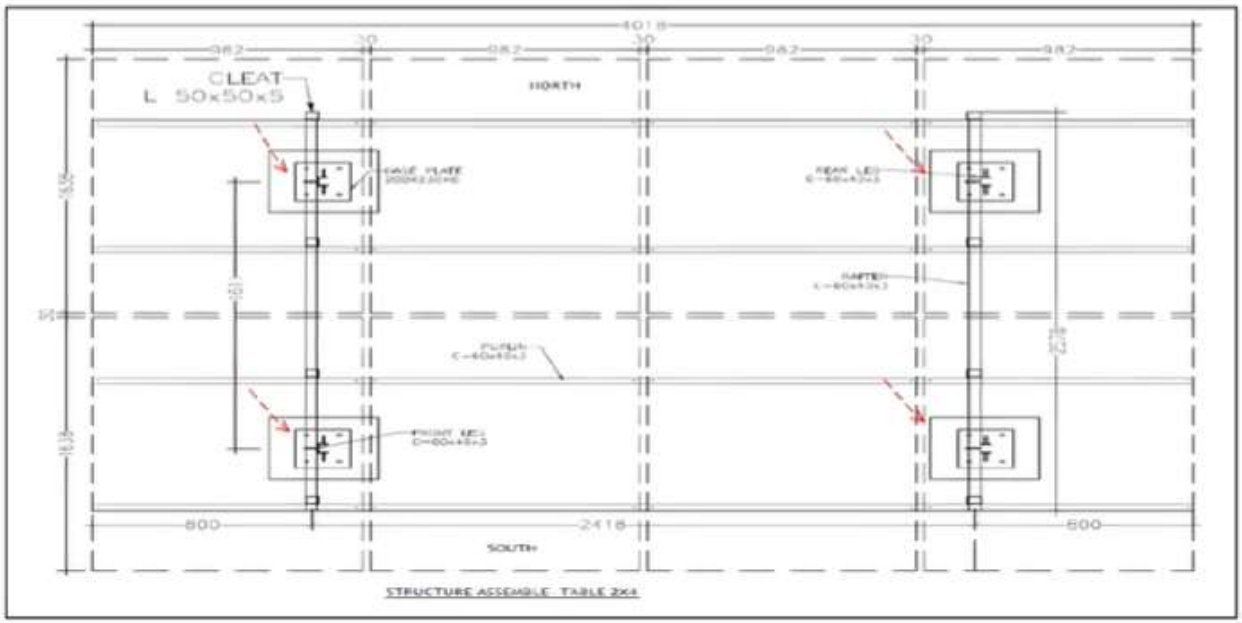
- Detailed drawings of structure, earthworks, falls, drainage etc.

CAD Simulation

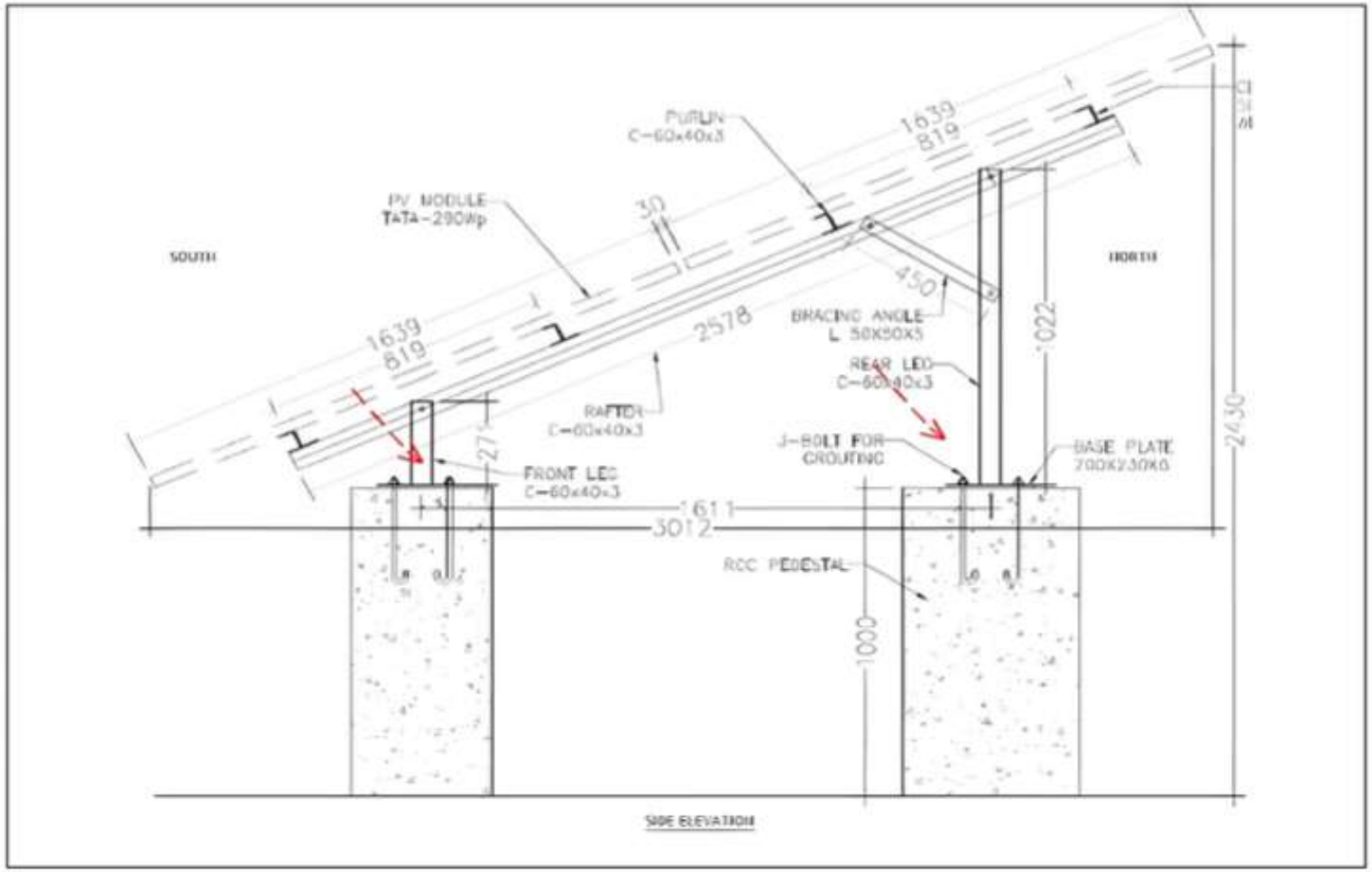
- Auto CAD (Computer Aided Design) Simulation, allows you to draw and edit 2D and 3D designs .
- They are used in Architecture and Design.



Civil/Structure Drawing



CAD Simulation





Let's Quantify,

SAMPLE BILL OF MATERIAL				
CIVIL STRUCTURAL COMPONENTS				
SL.	ITEM	DESCRIPTION	QTY	
1	Purlin	C-60X40X3(mm);LENGTH 4018mm	16100	mm
2	Rafter	C-60X40X3(mm);LENGTH 2578 mm	5200	mm
3	Bracing Angle	L-50X50X5 (mm); LENGTH 450mm	900	mm
4	Front Leg	C-60X40X3(mm);LENGTH 271 mm	600	mm
5	Rear Leg	C-60X40X3(mm);LENGTH 1022 mm	2100	mm
6	Cleat	50X50X5 (mm)	As Required	
7	Base Plate	200X230X6 (mm)	4	Nos.
8	J Type Bolt		16	Nos.
9	Bolt	M10	As Required	

4.6: Procurement of Solar PV system Component

4.6.1 Steps for Procurement of Solar PV System Components



Steps for Procurement of Solar PV System

Step 1: Approach Stores of the Company with BOM and Specifications.

Step 2: Identify and List the Variation in Equipment Specifications.

Step 3: Arrange the tools and consumables.

Step 4: List Statutory Requirements.

Step 5: Make sure all the Material Certificates are available with the supplier.

Step 6: Complete documentation regarding Procurement.

Step 1: Approach Stores of the Company with BOM and Specifications



You will procure these items:

Step 1

Solar Panel

Inverter

Charge Controller

Solar Cables

Solar Batteries

AC Distribution Board (ACDB)

DC Distribution Board (DCDB)

Step 1

Rated Power at Standard Test Conditions (STC)	Refers to the maximum electrical output that a solar panel can generate under controlled laboratory conditions.
Rated Power Tolerance	Refers to the allowable variation in the maximum power output of a solar panel compared to its specified value under STC.
Temperature Coefficient	Refers to the impact of temperature on the electrical output of a solar panel.
Open Circuit Voltage	It is the maximum voltage a solar panel can produce when there's no external load connected

Step 1

Voltage at MPP (Maximum Power Point)	Refers to the voltage at which solar panel operates to deliver maximum electrical power output.
Short Circuit Current	Refers to the maximum electrical current a panel can produce when its positive and negative terminals are directly shorted together.
Current at MPP	Refers to the voltage at which a solar panel operates to deliver its maximum electrical power output.
Efficiency	Refers to the percentage of solar energy (sunlight) that a solar panel can convert into usable

Step 1

Module Dimensions	Refer to the physical size and shape of the solar panel.
Operating Temperature	Refers to the actual temperature of the solar cells within the panel during electricity generation

Step 1

Solar PV Panel



SPECIFICATION SHEET										
Max Power Pmp (W)	10W	20W	40W	50W	75W	80W	100W	125W	150W	250W
Power Tolerence (+/-)	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Max Power Voltage Vmp (V)	16.85	16.95	17.15	17.25	16.92	17	18	18.15	18.25	30.72
Max Power Current Imp (A)	0.59	1.18	2.33	2.9	4.43	4.71	5.56	6.89	8.22	8.14
Open Circuit Voltage Voc (V)	20.9	21	21.2	21.3	21.82	22.18	22.3	22.4	22.5	37.8
Short Circuit Current Isc (A)	0.65	1.29	2.55	3.17	4.92	5.11	6.1	7.4	8.85	8.63
Max. System Voltage VDC	600	600	600	600	600	600	1000/600	1000/600	1000/600	1000
Pm Temperature Coefficient	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.43
Isc Temperature Coefficient (mA/K)	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	0.04
Voc Temp Coefficient (mV/K)	-2	-2	-2	-2	-2	-2	-2	-2	-2	-0.32
Nominal Operating cell temp (Celcius)	45	45	45	45	45	45	45	45	45	45

Inverter Specification

Step 1



Step 1

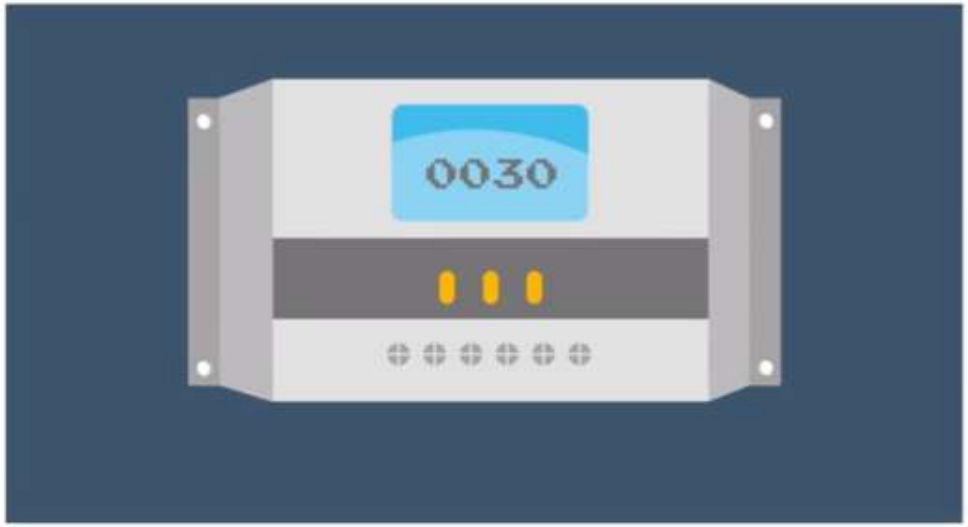
Choose the correct type of Inverter	There are three types of inverter which are String Inverters, Microinverters and Central Inverters. Choose as per the requirement.
AC Output Capacity	The AC output capacity of an inverter in a solar PV system refers to the maximum amount of AC electricity it can convert from DC power.
DC Input Voltage Range	Specifies the range of DC voltages the inverter can accept from the connected solar panels and function properly.
Operating temperature range	Refers to the range of ambient temperatures within which the inverter can function safely and efficiently

Step 1

SPECIFICATION SHEET						
Model	100W/200W/300W/400W/12 V	500W/24V	750W/24V	1000W/24 V	1500W/48V	2000W/48V
Rated Power	100W/200W/300W/400W	500W	750W	1000W	1500W	2000W
INPUT						
Nominal Input Voltage	12VDC	24VDC	24VDC	24VDC	48VDC	48VDC
Input Voltage Range (DC)	10.5-15.5VDC	21.0-31.0VDC	21.0-31VDC	42.0-62.0VDC	42.0-62.0VDC	42.0-62.0VD C
Battery Low Alarm	10.7 VDC	21.5VDC	21.5VDC	21.5VDC	43.0VDC	43.0VDC
Battery Low Shutdown	10.5 VDC	21.0VDC	21.0VDC	21.0VDC	42.0VDC	42.0VD C
Output						
Efficiency	85%					
Output Voltage	110VAC 50- Hz/220 VAC 50 Hz					
Voltage Regulation	+/-3% TO +/-10% RMS					
Output Waveform	Sine Wave form					
THD	<5%					
Overload	Above 110%					
Protection	Overload, Short Circuit, Reverse Polarity through DC fuse, over temperature					
General						
Operating Temp Range	0 deg C to 40 deg C					
Storage Temp Range	-20 deg C to 70 deg C					
Thermal Management/Cooling	Controlled Forced air cooling					
Relative Humidity	0-95% Non-Condensing					
Dimesions WxDxH (in mm)	200x361x85		240x380x100		250x450x125	250x450x125
Weight (kgs)	3.8		5.4	6	8	9
Indications	Low battery, Overload/ Short circuit, Inverter on					

Charge Controller Specification

Step 1



Types of Charge Controller

Step 1

1. Pulse with Modulation (PWM) Solar Charge Controller:

The PWM charge controller is a good low-cost solution for small systems only, when solar cell temperature is moderate to high (between 45°C and 75°C)



Step 1

TECHNICAL SPECIFICATION	
Type	Series Regulators Common Negative
Technology	Microcontroller Based Control
System Voltage	240V
ELECTRICAL PARAMETERS	
Charging current	40 Amp
Solar Array	Single Array
Bulk Voltage	282 +/- 2V
	Adjustable 270 – 290 V
Absorption period	Hold battery voltage or bulk setting for a cumulative period of 1 hr
Float Voltage	270 +/- 2V
Indications	LED indications for Bulk Mode, Float Mode, High Current, Low Battery
	High Battery, Array/ Battery Reverse Polarity
	Analog type voltmeter and ammeter with a selector
	switch for monitoring voltage of solar array/ battery and charge current
ENVIORNMENTAL PARAMETERS	
Operating Temperature	0 Deg C to + 40 Deg C
Storage Temperature	0 Deg C to + 55 Deg C
Relative Humidity	0– 95% Non Condensing
Dimensions WxDxH (in mm)	400 x 475 x 151
Weight (in Kgs)	9.5

Types of Charge Controller

Step 1

2. Maximum Power Point Tracking (MPPT) Solar Charge Controller

- The MPPT are most common these days and can provide a of 30% more power than the PWM controllers.
- The MPPT controllers allow the strings of panels to be connected in series for higher voltages, keeping the amperage lower and the wire size smaller, especially for long-wire runs to the PV array.



Step 1

TECHNICAL SPECIFICATION	
Type of Product	MPPT With Load Charger
Voltage(V)	12 V - 24 V
Current(A)	20 A
Application	Off-grid and micro grid. Telecom grid solar system. Home lighting system and street lighting system.
Product Warranty	2 Years

Solar Cables

Step 1

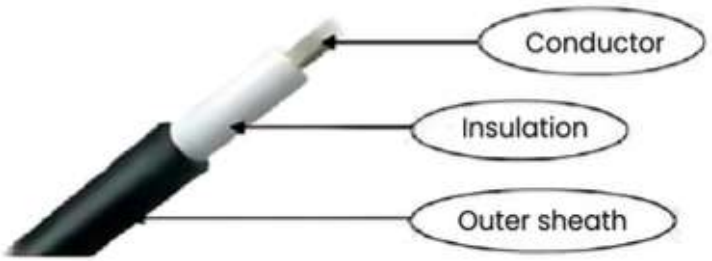


Step 1

Solar Cable:

- Solar cable interconnects solar panels and other electrical components of a photovoltaic system.
- Solar cables are designed to be UV-resistant and weather-resistant.
- They can be used within a large temperature range and are generally laid outside.





Measurement Specifications

Step 1

Cable Specifications

- Generally sold in 6, 30, 50 and 100 foot.
- Wire Gauze size : 30 or 25 Amp capacity.
- Can handle either 600 or 1000 volts.

Solar Batteries

Step 1



Step 1

Solar Batteries

Deep-cycle, lead-acid batteries are widely used in renewable energy and grid-backup system and are ideally suited for these applications because of their long, reliable life and low cost of ownership.



Lead-acid Battery



Sealed Lead-acid

Step 1

Capacity

Battery capacity is a measure of the amount of energy stored in the battery.

Voltage

The battery bank voltage must be considered to ensure it matches the system requirements. It is often determined by the inverter specifications if installing a DC-to-AC system or by the voltage of the loads in a DC system

Cycle Life

The life cycle is crucial to consider. It provides the number of discharge cycles that a battery can provide before capacity drops to a specified percentage of rated capacity.

Step 1



Batteries from different manufacturers may have the same capacity and energy content and be similar in weight. But the design, materials, process and quality will influence how long the battery will cycle

SPECIFICATION SHEET

Model	Nominal Voltage (V)	Nominal Capacity (AH)	Dimensions (mm)				Approx Weight (Kg)	Terminal Position	Terminal Type
			Length	Width	Height	Total Height			
Vendor Specific	2	100	103	206	355	409	12.6	G	M10
	2	150	103	206	355	409	15.3	G	
	2	200	103	206	355	409	17.2	G	
	2	250	124	206	355	409	20.8	G	
	2	300	145	206	355	409	24.3	G	
	2	350	124	206	470	525	26.9	G	
	2	420	145	206	470	525	31.5	G	
	2	500	166	206	470	525	36.1	G	

Step 2: Identify and List the Variation in Equipment Specifications



Specification for Variations

Step 2

Ensure

- That the components which are being ordered are matching design requirements.
- Get the list of variations from the vendor itself that has proper technical justification for further evaluation by design team
- Obtain the approval of the revised drawing

Step 3: Arrange the Tools and Consumables



Step 3



Solar path finder



Compass



Angle finder



Torpedo Level



Chalk Line



Hole Saw



Torque Wrench



Wire Strippers



Cable Cutter



Caulking Gun

Step 3



A – hammer

B – plier

C – screw driver

d – measuring tape

E – cordless grill

F – multimeter

G – utility knife

H – ratchet set

Step 3



Electrical Tape



Cable Ties



Cable Clips



Silicon Caulking



Nuts, Bolts and Fasteners



Splice



Washer



Mounting clips

List of Consumables

Step 3





Screw and Grooves

Step 4: List Statutory Requirements



Step 4

MNRE (Ministry of New & Renewable Energy)	• To obtain subsidies or participate in government schemes, registration of module types with the Indian (MNRE) is required. • Only MNRE approved suppliers and modules types are entitled to the MNRE policy schemes & subsidies.
Bureau of Indian Standards (BIS)	Major BIS standard for Solar PV modules are: • IS 14286 (adopts IEC 61215) • IS 61701
System Warranty	Current market guarantee is of 25 years warranty. Which means that the guaranteed electrical production is of 10 years at 90% of rated power output and 25 years at 80%.

Step 5: Make sure all the Material Certificates are Available with Supplier



Step 6: Complete Documentation Regarding Procurement





4.7: Verification of Components on-site

4.7.1 Plan and Receive the Equipment at Site





Be prepared with the Pre-installation work before the components arrive at the site!

Handling of the Material



Handling of the Material includes:



Transport



Storage



Movement of Material



Operations



Cleaning

Transportation of Solar Panels



Transport Truck



Proper Packaging

During transportation, specialized vehicles and packaging materials safeguard panels against damage en route from manufacturing facilities to the customer's site.

Storage of Panels

The panels must be stored in controlled environments to shield them from potential hazards like moisture and extreme temperatures.



Handling Panels during Installation

During installation, trained personnel utilise precise handling techniques to prevent scratches or cracks in the delicate photovoltaic cells.



Maintenance of the Panels

Gentle cleaning to remove dust and debris and ensure optimal performance that prolongs the panels' lifespan.









- Take swift action to minimise downtime.



- Facilitate quick replacements with the help of stocked spare panels



- Adhere to strict safety protocols.



Do's

Don'ts



1. Open the boxes carefully. Wrap the unpack box properly once the inspection is done. Follow Vendor provided storage procedure strictly.

2. Always use gloves while handling hazard potential equipment like batteries.

3. Wrap the nut & bolts properly.

4. Always read user manual before opening the pallet.

5. Always take out the PV module from pallet one by one.

6. Handle PV modules with care, avoiding excessive bending or flexing.



Do's

7. Use Appropriate lifting equipment and techniques to prevent damage.

8. Store PV modules in a dry, clean area away from potential damage sources.

9. Inspect PV modules for any visible damage or defects before installation.

10. Follow manufacturer's guidelines for handling and installation.

11. Wear appropriate personal protective equipment (PPE) when handling PV materials.

12. Secure PV modules during transport to prevent shifting or sliding.

1. Do not spread the parts/ materials on surface.

2. Never handle the equipment like batteries with bare hands as it contains acid.

3. Do not start using tools & tackles without proper knowledge.

4. Do not handle the PV module alone. There should always be minimum two person for handing the solar PV module.

5. Do not stack more than ten modules on each other.

6. Do not drop or mishandle PV modules.



7. Avoid walking on PV modules or applying excessive pressure.

8. Do not stack PV modules too high, risking collapse damage.

9. Do not install cracked or damaged PV modules.

10. Avoid using sharp or abrasive materials near PV modules.

11. Do not expose PV modules to extreme temperatures or weather conditions.

12. Avoid twisting or torsional forces when handling PV modules.



Check Materials Received On- Site



Sl. no	Item to be checked/ verified after receiving the purchased components at Site.	Yes/ No	Remarks
1	Check Component qty as per final BOM		
2	Check Equipment and parts are properly labeled as required		
3	Make ensure that all the components received damaged free		
4	Check the all the require documents like Manufacturer operation and maintenance instructions, Manufacturers As built drawing, Catalogues, storage & handling procedure, list of deviation are received		
5	System warranty		
6	All QC inspection reports		
7	Certificates regulatory required as per local/ national		
8	Check the received accessories for mounting panels are compatible with the fabricated structures at site		
9	Check all the recommended tools & tackles are received		